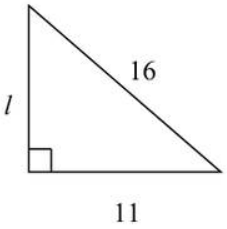
Type MS · Getting one exact trig ratio from another

Answer 1 of 4 · Question 2 part a · 4 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Number		
2 (a)	$\cos ABC = \dfrac{(2x)^2+(4x)^2-(3x)^2}{2\times 2x\times 4x} = \dfrac{x^2(4+16-9)}{x^2(16)} = \dfrac{11}{16}$	M1A1
	 $l = \sqrt{16^2-11^2} = 3\sqrt{15}$	M1
	$\sin ABC = \dfrac{3\sqrt{15}}{16}$ *	A1 [4]
	ALT	
	$\sin^2 ABC = 1 - \dfrac{121}{256} = \dfrac{135}{256}$ ⊳ $\sin ABC = \dfrac{3\sqrt{15}}{16}$ *	{M1A1}
(b)	$\dfrac{75\sqrt{15}}{64} = \dfrac{1}{2}\times 2x\times 4x\times \dfrac{3\sqrt{15}}{16} \Rightarrow x^2 = \dfrac{25}{16} \Rightarrow x = \dfrac{5}{4}$ oe (positive root only)	M1A1 [2]
		Total 6 marks

Type MS · Getting one exact trig ratio from another

Answer 1 of 4 · Question 2 part a · 4 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Getting one exact trig ratio from another

Answer 2 of 4 · Question 2 part a.i · 0 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part a.i in this mark scheme extract

M1	Uses b^2-4ac on the given quadratic equation with correct a, b and c; $a=3(k+2)$ $b=k+5$ $c=k$ **and** a correct substitution to obtain $(k+5)^2-4\times3\times(k+2)(k)$ **Note:** Accept for this mark any inequality, equals sign and even b^2-4ac used on its own.
M1	For attempting to expand the brackets and form a 3TQ in terms of k. Allow as a minimum at least one term correct. $k^2+10k+25-12k^2-24k \Rightarrow (-11k^2-14k+25)$ M0M1 is possible here.
A1	For the correct 3TQ with the correct inequality. **Note:** Allow > or < in place of $\geqslant$ and $\leqslant$ for this mark $-11k^2-14k+25 \geqslant 0$ or $11k^2+14k-25 \leqslant 0$
M1	For an attempt to solve their 3TQ, (provided it is a 3TQ) in terms of k by any acceptable method. See General Guidance for the definition of an attempt by factorisation, formula or completing the square. Use of calculators: if their 3TQ is incorrect, do not award this mark if working is not seen. $(11k+25)(k-1)=0 \Rightarrow k=1,\ -\frac{25}{11}$
M1	For forming the correct inequality with **their** critical values, provided they have been obtained from a 3TQ, must be a closed region. $\left(-\frac{25}{11} \leqslant k \leqslant 1\right)$ ft their values from their $-11k^2+14k-25 \geqslant 0$ or $11k^2-14k+25 \leqslant 0$
A1	For the correct inequality. $-\frac{25}{11} \leqslant k \leqslant 1$

Type MS · Getting one exact trig ratio from another

Answer 2 of 4 · Question 2 part a.i · 0 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part a.i in this mark scheme extract

Question number	Scheme	Marks
2 (a)(i)	$(\tan\alpha =)\frac{4}{3}$	B1
(a)(ii)	$(\tan\beta =)-\frac{1}{\sqrt{3}}\quad\left(=-\frac{\sqrt{3}}{3}\right)$	B1 B1 (3)
(b)	$\tan(\alpha+\beta)=\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}$	
	$=\frac{\frac{4}{3}+\left(-\frac{1}{\sqrt{3}}\right)}{1-\left(\frac{4}{3}\right)\left(-\frac{1}{\sqrt{3}}\right)}$	M1
	$=\frac{\frac{4\sqrt{3}\pm 3}{3\sqrt{3}}}{\frac{3\sqrt{3}\pm 4}{3\sqrt{3}}}$ or $\frac{\frac{12+\left(-3\sqrt{3}\right)}{9}}{\frac{9+4\sqrt{3}}{9}}$	dM1
	$=\frac{4\sqrt{3}-3}{3\sqrt{3}+4}$	A1 cso (3)
		Total 6 marks

Type MS · Getting one exact trig ratio from another

Answer 2 of 4 · Question 2 part a.i · 0 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part a.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Getting one exact trig ratio from another

Answer 3 of 4 · Question 2 part a.ii · 3 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part a.ii in this mark scheme extract

M1	Uses b^2-4ac on the given quadratic equation with correct a, b and c; $a=3(k+2)$ $b=k+5$ $c=k$ **and** a correct substitution to obtain $(k+5)^2-4\times3\times(k+2)(k)$ **Note:** Accept for this mark any inequality, equals sign and even b^2-4ac used on its own.
M1	For attempting to expand the brackets and form a 3TQ in terms of k. Allow as a minimum at least one term correct. $k^2+10k+25-12k^2-24k \Rightarrow (-11k^2-14k+25)$ M0M1 is possible here.
A1	For the correct 3TQ with the correct inequality. **Note:** Allow > or < in place of $\geqslant$ and $\leqslant$ for this mark $-11k^2-14k+25 \geqslant 0$ or $11k^2+14k-25 \leqslant 0$
M1	For an attempt to solve their 3TQ, (provided it is a 3TQ) in terms of k by any acceptable method. See General Guidance for the definition of an attempt by factorisation, formula or completing the square. Use of calculators: if their 3TQ is incorrect, do not award this mark if working is not seen. $(11k+25)(k-1)=0 \Rightarrow k=1,\ -\frac{25}{11}$
M1	For forming the correct inequality with **their** critical values, provided they have been obtained from a 3TQ, must be a closed region. $\left(-\frac{25}{11} \leqslant k \leqslant 1\right)$ ft their values from their $-11k^2+14k-25 \geqslant 0$ or $11k^2-14k+25 \leqslant 0$
A1	For the correct inequality. $-\frac{25}{11} \leqslant k \leqslant 1$

Type MS · Getting one exact trig ratio from another

Answer 3 of 4 · Question 2 part a.ii · 3 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part a.ii in this mark scheme extract

Question number	Scheme	Marks
2 (a)(i)	$(\tan\alpha =)\frac{4}{3}$	B1
(a)(ii)	$(\tan\beta =)-\frac{1}{\sqrt{3}}\quad\left(=-\frac{\sqrt{3}}{3}\right)$	B1 B1 (3)
(b)	$\tan(\alpha+\beta)=\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}$	
	$=\frac{\frac{4}{3}+\left(-\frac{1}{\sqrt{3}}\right)}{1-\left(\frac{4}{3}\right)\left(-\frac{1}{\sqrt{3}}\right)}$	M1
	$=\frac{\frac{4\sqrt{3}\pm 3}{3\sqrt{3}}}{\frac{3\sqrt{3}\pm 4}{3\sqrt{3}}}$ or $\frac{12+\left(-3\sqrt{3}\right)}{\frac{9+4\sqrt{3}}{9}}$	dM1
	$=\frac{4\sqrt{3}-3}{3\sqrt{3}+4}$	A1 cso (3)
		Total 6 marks

Type MS · Getting one exact trig ratio from another

Answer 3 of 4 · Question 2 part a.ii · 3 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part a.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Getting one exact trig ratio from another

Answer 4 of 4 · Question 6 part a.ii · 5 marks

2022 JAN · Paper 01 · extract 1 of 4 · find part a.ii in this mark scheme extract

6(a)(i)	$\frac{\sin 30^\circ}{x} = \frac{\sin ACB}{x+3}$ $\sin\theta^\circ = \frac{x+3}{2x}$ *	M1 A1 cso
(ii)	$\cos^2\theta^\circ = 1-\left(\frac{x+3}{2x}\right)^2$ $\cos^2\theta^\circ = \frac{(2x)^2-(x+3)^2}{(2x)^2}$ $\cos\theta^\circ = \frac{\sqrt{3x^2-6x-9}}{2x}$ *	M1 M1 A1 [5]
Alt – use of right-angled triangle with Pythagoras' theorem		
	Adjacent $= \sqrt{(2x)^2-(x+3)^2}$ $\cos\theta = \frac{\sqrt{(2x)^2-(x+3)^2}}{2x}$ $\cos\theta^\circ = \frac{\sqrt{3x^2-6x-9}}{2x}$ *	[M1 M1 A1]
(b)	$\frac{\angle BAC}{30^\circ} = \frac{7}{2} \Rightarrow \angle BAC = 105^\circ$ $\theta = 180-30-105 = 45$ $\cos 45^o = \frac{\sqrt{2}}{2} = \frac{\sqrt{3x^2-6x-9}}{2x} \Rightarrow 2x^2 = 3x^2-6x-9 \Rightarrow x^2-6x-9=0$ $x^2-6x-9=0 \Rightarrow (x-3)^2-18=0 \Rightarrow x=\ldots$ $x = 3+3\sqrt{2}$	B1 B1 M1 M1 A1 [5]
Alt – last three marks		
	$\sin\theta^\circ = \frac{x+3}{2x} = \frac{\sqrt{2}}{2} \Rightarrow x+3 = \sqrt{2}x$ $x+3=\sqrt{2}x \Rightarrow x\left(\sqrt{2}-1\right)=3 \Rightarrow x = \frac{3}{\sqrt{2}-1}$ $x = 3+3\sqrt{2}$	[M1 M1 A1]
		Total 10 marks

Type MS · Getting one exact trig ratio from another

Answer 4 of 4 · Question 6 part a.ii · 5 marks

2022 JAN · Paper 01 · extract 2 of 4 · find part a.ii in this mark scheme extract

Part	Marks	Scheme
(a) (i)	M1	For using a correct sine rule to give, $\frac{\sin 30^\circ}{x} = \frac{\sin ACB}{x+3}$
	A1 cso	For correctly obtaining the expression for $\sin\theta$ $\sin\theta^\circ = \frac{x+3}{2x}$ *
(ii)	M1	For using the Pythagorean identity $\cos^2\theta^\circ = 1-\left(\frac{x+3}{2x}\right)^2$
	M1	For simplifying to form a single fraction $\cos^2\theta^\circ = \frac{(2x)^2-(x+3)^2}{(2x)^2}$
	A1 cso	For simplifying to achieve the given expression, $\cos\theta^\circ = \frac{\sqrt{3x^2-6x-9}}{2x}$ * Note this is a show question
Alt – use of right-angled triangle with Pythagoras' theorem		
	M1	For use of a right-angled triangle with Pythagoras' theorem to determine the adjacent Adjacent $= \sqrt{(2x)^2-(x+3)^2}$
	M1	For use of cosine ratio $\cos\theta = \frac{\sqrt{(2x)^2-(x+3)^2}}{2x}$
	A1 cso	For simplifying to achieve the given expression, $\cos\theta^\circ = \frac{\sqrt{3x^2-6x-9}}{2x}$ * Note this is a show question
(b)	B1	For finding the size of $\angle BAC$ $\frac{\angle BAC}{30^\circ} = \frac{7}{2} \Rightarrow \angle BAC = 105^\circ$
	B1	For finding the value of $\theta = 180-30-105 = 45$
	M1	For substituting the value of $\angle ABC$ into the given expression for $\cos\theta$ and forming a 3TQ, condone arithmetic errors in rearrangement. $\cos 45^o = \frac{\sqrt{2}}{2} = \frac{\sqrt{3x^2-6x-9}}{2x} \Rightarrow 2x^2 = 3x^2-6x-9 \Rightarrow x^2-6x-9=0$ Allow for use of their 45° but this must come from an attempt at working with the ratio. Do not allow if their 45° is 30°.
	M1	For an attempt to solve their 3TQ by any valid method (see general guidance) $x^2-6x-9=0 \Rightarrow (x-3)^2-18=0 \Rightarrow x=...$
	A1 cao	For the correct value of x in the correct form $x = 3+3\sqrt{2}$ Allow $a = 3, b = 2$
Alternative method		
	M1	For substituting the value of $\angle ABC$ into the given expression for $\sin\theta$ and forming a linear equation $\sin\theta^\circ = \frac{x+3}{2x} = \frac{\sqrt{2}}{2} \Rightarrow x+3 = \sqrt{2}x$

Type MS · Getting one exact trig ratio from another

Answer 4 of 4 · Question 6 part a.ii · 5 marks

2022 JAN · Paper 01 · extract 3 of 4 · find part a.ii in this mark scheme extract

	M1	For an attempt to solve their linear equation $x+3=\sqrt{2}x \Rightarrow x\left(\sqrt{2}-1\right)=3 \Rightarrow x=\dfrac{3}{\sqrt{2}-1}$
	A1 **cao**	For the correct value of x in the correct form $\quad x=3+3\sqrt{2}$ Allow $a = 3, b = 2$

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Type MS · Getting one exact trig ratio from another

Answer 4 of 4 · Question 6 part a.ii · 5 marks

2022 JAN · Paper 01 · extract 4 of 4 · find part a.ii in this mark scheme extract

Question	Scheme	Marks